



COURSE SLIDES

AI Vibe Coding for Algorithmic Trading

Course Code: C728

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

Meet your trainer

1

Trainer

Dr. Alfred Ang

2

Your experience

Share a Python or finance project.

3

Your goal

Name one trading workflow to improve.

4

Your responsibility

Review assumptions and verify results.

Let's Know Each Other

1 Your name and organisation / role.

2 Your experience with this subject (if any).

3 What you want to be able to do after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Your course toolkit

1

Slides

Visual concepts and worked examples.

2

Learner Guide

Detailed lab procedures and troubleshooting.

3

Lab folders

Sample Python scripts and prompt PDFs.

4

Shared package

Use the course Drive folder and GitHub README.

SCHEDULE

Lesson Plan — 2 Days, 15 instructional hours

Day 1

- Day 1 — From prompt to validated signal
 - Topic 1: Getting Started with AI Vibe Coding for Trading (Labs 1–2)
 - Topic 2: Analysing Markets with AI (Labs 3–4)

Day 2

- Day 2 — From backtest to monitored paper bot
 - Topic 3: Building and Backtesting Strategies with AI (Labs 5–6)
 - Topic 4: Automating and Running Trading with AI (Labs 7–8)
- Daily timing
 - 09:00–18:00 · 1-hour lunch · 30 minutes tea breaks

Learning Outcomes

1

Prompt and review

Specify, generate and validate Python trading code with an AI assistant.

2

Data and signals

Validate price data, visualise indicators and construct causal signals.

3

Backtest and risk

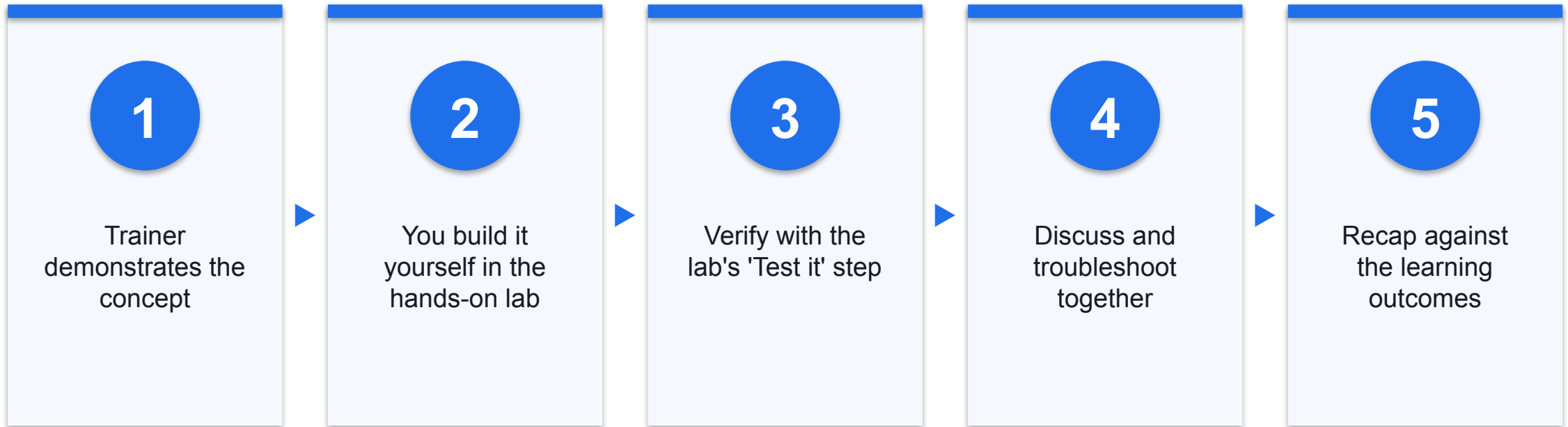
Backtest with execution delay and costs, and interpret return and risk.

4

Validate and monitor

Validate parameters chronologically and operate a monitored paper-trading workflow.

How You'll Learn





CORE CONCEPTS

Research before execution

Own the logic

1

Contract

State data, timing and acceptance checks.

2

Evidence

Run checks and inspect charts.

3

Risk

Model costs and exposure limits.

4

Control

Paper mode and observable failures.

The AI review loop

1

Specify

Describe the smallest useful change.

2

Generate

Ask for a plan and focused code.

3

Inspect

Read assumptions and data access.

4

Verify

Run known-result tests before accepting.

RESEARCH DISCIPLINE

A plausible chart is not proof.

Demand causal signals, reproducible inputs and visible failure handling.

How Every Lab Progresses

1

Describe → generate → review → run → compare → refine

TOPIC 01

Getting Started with AI Vibe Coding for Trading

Python environment · prompt contracts · data ingestion

KEY CONCEPTS

Key Concepts — Getting Started with AI Vibe Coding for Trading

1

A trading rule must define instrument, timing, exposure and exit.

2

AI generates candidates; humans own assumptions, tests and risk.

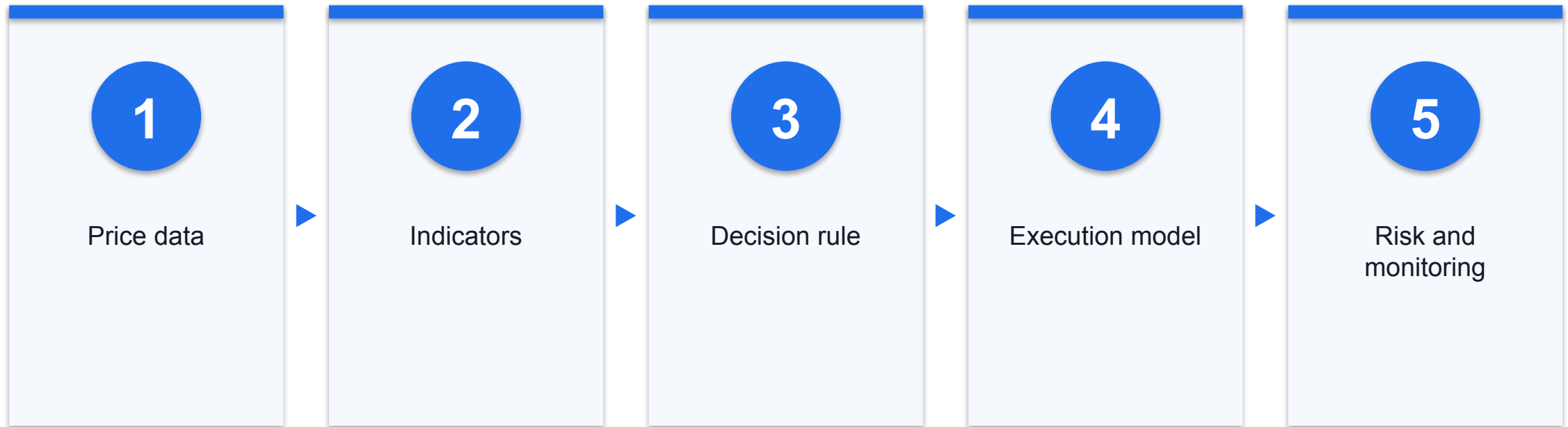
3

A prompt contract states inputs, outputs, constraints and acceptance checks.

4

Keep secrets and account data out of prompts and Git.

Algorithmic trading: the system



The strategy specification

1

Universe

One synthetic daily series; long or flat.

2

Entry / exit

Fast SMA above slow SMA / otherwise flat.

3

Timing

Observe close; lag exposure one bar.

4

Costs

10 bps per unit of exposure turnover.

From vague request to prompt contract

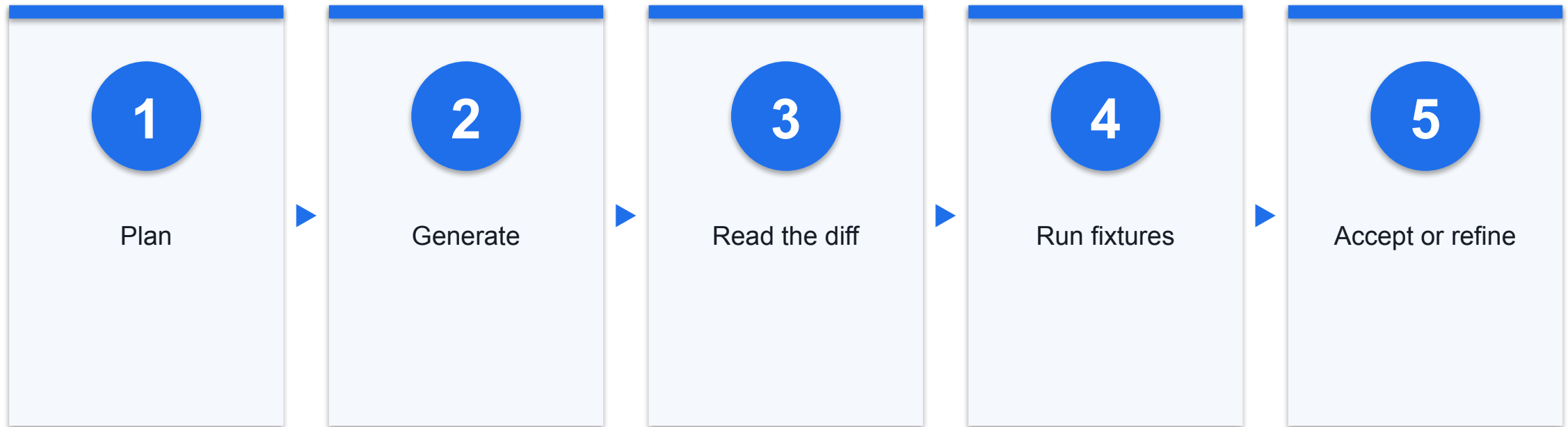
VAGUE

- Build a profitable trading bot
 - Missing data, timing and failure criteria

REVIEWABLE

- Build a reproducible SMA research pipeline
 - Specify schema, lag, costs and fixtures
 - Return a plan, focused diff and tests

Human control stays in the loop



Choose your assistant

1

Cursor

Use project context; inspect proposed edits.

2

GitHub Copilot

Discuss a focused task and review suggestions.

3

Claude

Provide a sanitised contract and minimal context.

4

Common discipline

Verify API claims against official docs.

Data is part of the strategy

1

Schema

date, close; finite positive price.

2

Provenance

Symbol, provider, timezone and adjustment policy.

3

Continuity

Missing sessions and duplicate dates.

4

Reproducibility

Keep a source snapshot and configuration.

WHAT YOU'LL DO

Hands-On Labs — Getting Started with AI Vibe Coding for Trading

Lab 1

- Environment and AI workflow

Lab 2

- Fetch and validate price data

Environment and AI workflow

LAB 1

Use AI to build and review environment and ai workflow; demonstrate the result with executable evidence.

You'll build

preview.csv

Tools: Python · AI coding assistant · synthetic dataset

Environment and AI workflow

Test it

756 observations and environment OK

Fetch and validate price data

LAB 2

Use AI to build and review fetch and validate price data; demonstrate the result with executable evidence.

You'll build

prices.csv

Tools: Python · AI coding assistant · synthetic dataset

Fetch and validate price data

Test it

both invalid fixtures rejected

TOPIC RECAP

Recap — Getting Started with AI Vibe Coding for Trading

1

You can now: LO1: Specify, generate and validate Python trading code with an AI assistant.

TOPIC 02

Analysing Markets with AI

Price data · indicators · signals · AI debugging

Key Concepts — Analysing Markets with AI

1

Validate ordering, duplicate dates, missing prices and positive values.

2

Rolling indicators use past and current observations only.

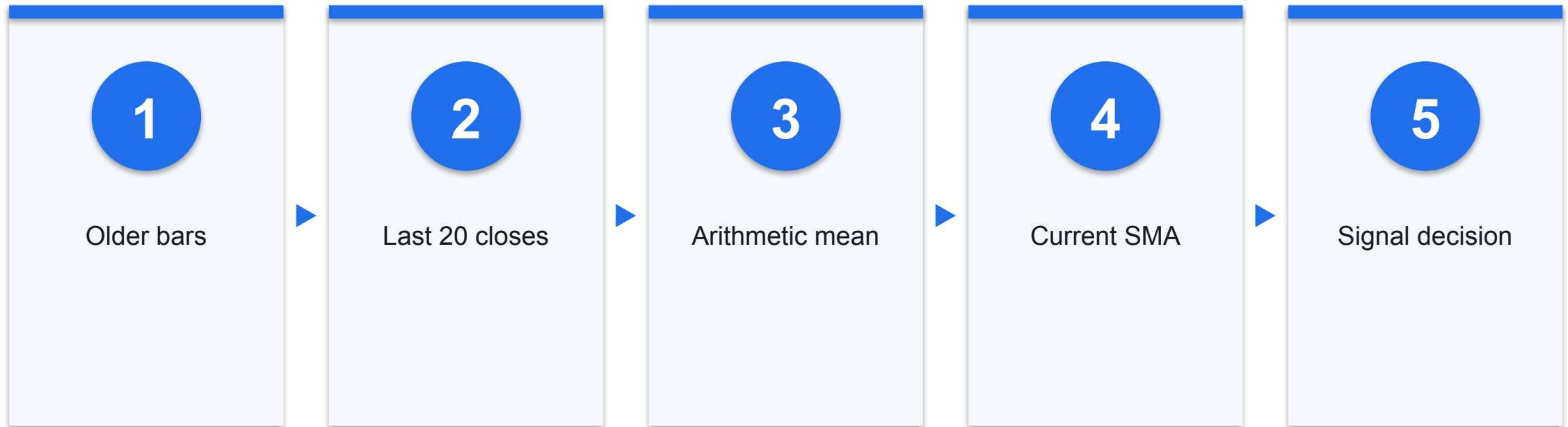
3

A signal is a decision; a position is exposure after execution delay.

4

Use charts and small examples to debug generated code.

A rolling indicator sees a window



Indicator families

1

SMA / EMA

Trend smoothing; EMA weights recent observations.

2

RSI

Relative gains/losses; specify smoothing convention.

3

Bollinger bands

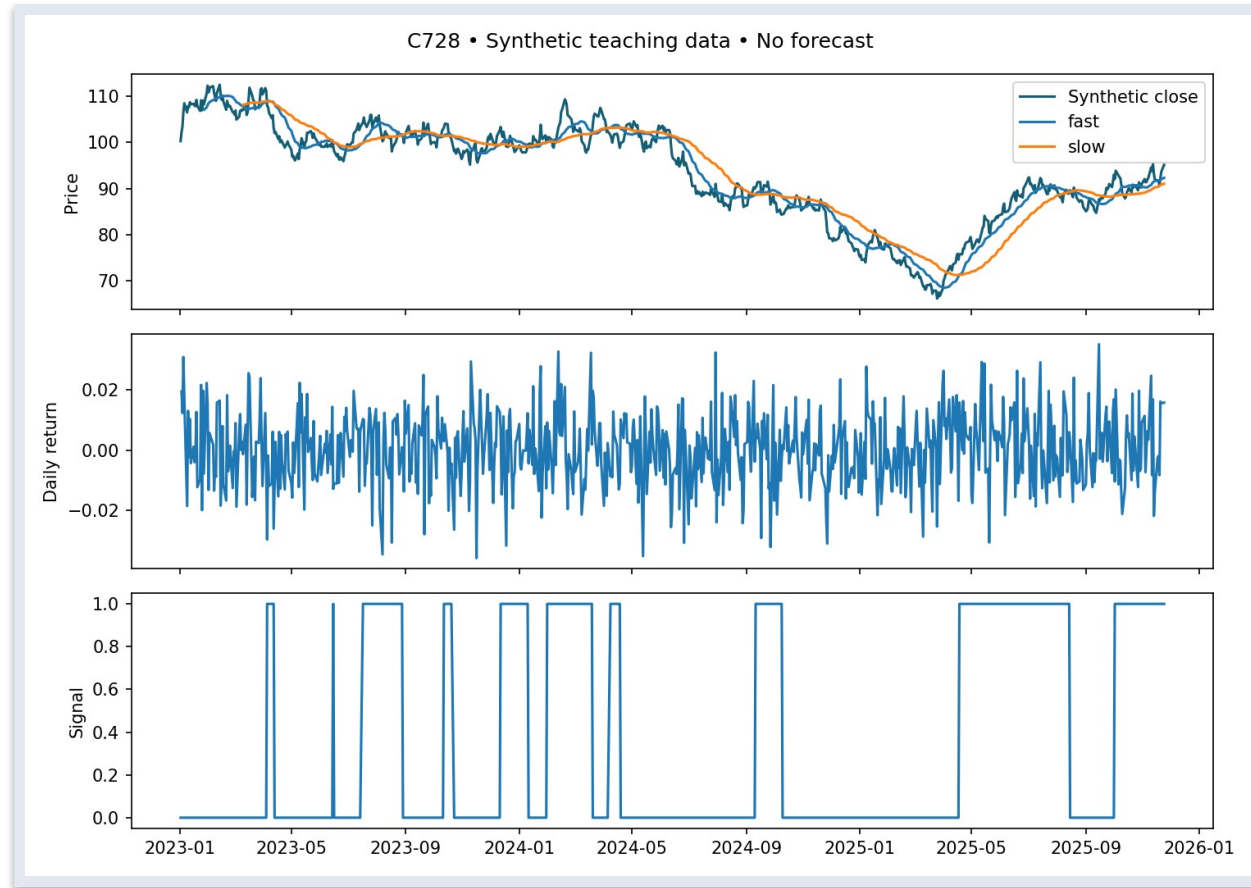
Moving mean $\pm$ volatility multiple.

4

MACD

Difference between fast and slow EMAs.

Price, moving averages and signal



Synthetic data; 20/50-bar means. Warm-up produces no position.

Signals are not positions

DECISION

- Signal at close t
 - Uses observations through t
 - Cannot earn the return already observed

EXPOSURE

- Position for next interval
 - $\text{position}[t] = \text{signal}[t-1]$
 - First observation remains flat

Debug with a small fixture

1

Warm-up

50-bar SMA first appears at row index 49.

2

Flat prices

Fast and slow means match; signal stays flat.

3

Future perturbation

Change future prices; earlier signals stay identical.

4

Minimal repair

Change one assumption; rerun accepted checks.

Hands-On Labs — Analysing Markets with AI

Lab 3

- Indicators and visual evidence

Lab 4

- Generate and debug trading signals

Indicators and visual evidence

LAB 3

Use AI to build and review indicators and visual evidence; demonstrate the result with executable evidence.

You'll build

indicators.csv and indicators.png

Tools: Python · AI coding assistant · synthetic dataset

Indicators and visual evidence

Test it

first valid 20-bar mean matches an independent calculation

Generate and debug trading signals

LAB 4

Use AI to build and review generate and debug trading signals; demonstrate the result with executable evidence.

You'll build

signals.csv

Tools: Python · AI coding assistant · synthetic dataset

Generate and debug trading signals

Test it

earlier signals unchanged after future-price perturbation

Recap — Analysing Markets with AI

1

You can now: LO2: Validate price data, visualise indicators and construct causal signals.

TOPIC 03

Building and Backtesting Strategies with AI

Strategy design · backtesting · performance · refactoring

KEY CONCEPTS

Key Concepts — Building and Backtesting Strategies with AI

1

A backtest is an execution model, not a promise of profit.

2

Lag close-derived signals by one bar for close-to-close return accounting.

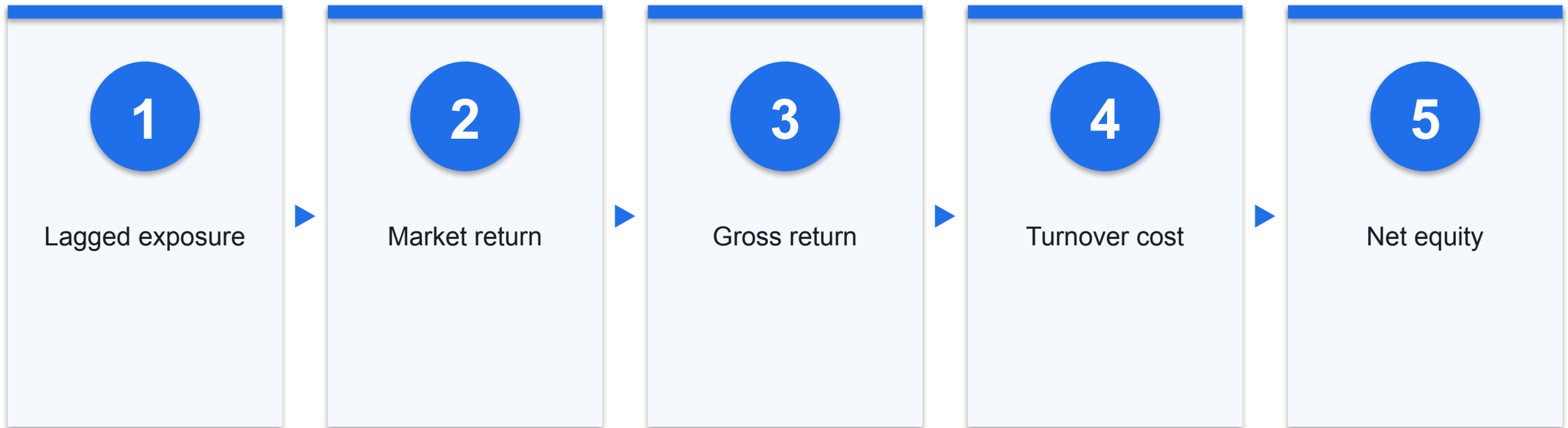
3

Turnover incurs transaction costs; compare with buy and hold.

4

Sharpe and drawdown describe different risks.

Backtest accounting



A hand-worked bar

1

Exposure

Yesterday signal = 1 $\rightarrow$ position = 1.

2

Market move

Close rises 100 $\rightarrow$ 102: return = 2%.

3

Cost

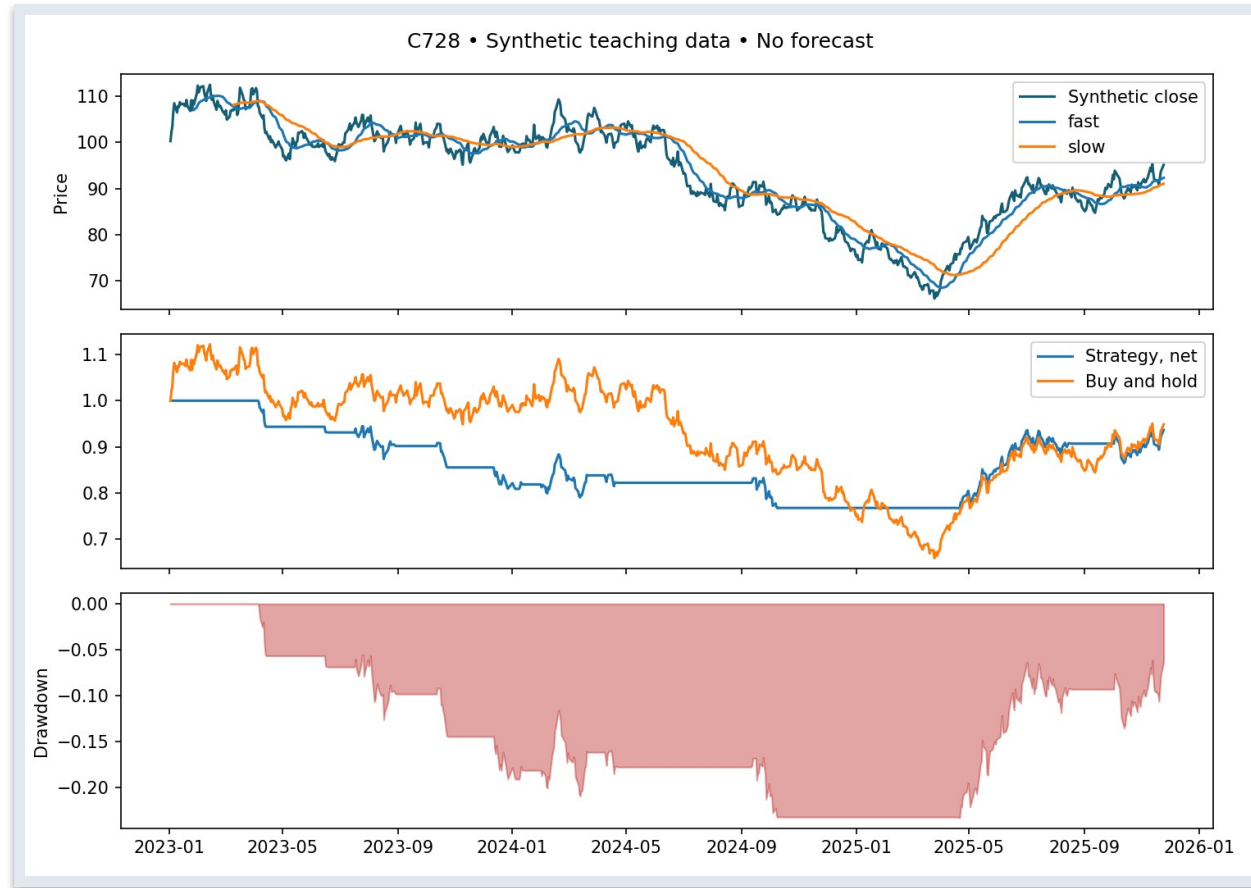
Entry turnover 1 $\times$ 10 bps = 0.1%.

4

Net result

2% - 0.1% = 1.9%; equity 1 $\rightarrow$ 1.019.

Strategy, benchmark and drawdown



Net of the stated turnover cost; simplified close-to-close execution.

Performance has several dimensions

1

Total return

Final equity / initial equity – 1.

2

Sharpe

Mean daily net return / sample standard deviation $\times \sqrt{252}$; risk-free rate = 0.

3

Maximum drawdown

Worst equity / running peak – 1.

4

Turnover

Sum of absolute exposure changes; reveals cost sensitivity.

Drawdown: a known-result fixture

1

Start

Equity = 1.00

2

Peak

Equity = 1.20

3

Trough

Equity = 0.90 → drawdown = -25%.

4

Recovery

Equity = 1.30 → new peak; current drawdown = 0%.

Review before refactoring

PRESERVE

- Behaviour contract
 - Same inputs → same signals and costs
 - Reject invalid dates and prices

IMPROVE

- Code improvement
 - Extract pure functions and explicit parameters
 - Preserve lag, warm-up and fixture results

What this backtest omits

1

Execution

Spread, gaps, partial fills and latency.

2

Data

Dividends, corporate actions and survivorship.

3

Portfolio

Financing, taxes and multi-asset constraints.

4

Meaning

A teaching simulation; no profitability forecast.

WHAT YOU'LL DO

Hands-On Labs — Building and Backtesting Strategies with AI

Lab 5

- Build a cost-aware backtest

Lab 6

- Performance, risk and code review

Build a cost-aware backtest

LAB 5

Use AI to create and review a cost-aware backtest; demonstrate the result with executable evidence.

You'll build

backtest.csv and backtest.png

Tools: Python · AI coding assistant · synthetic dataset

Build a cost-aware backtest

Test it

one-bar lag, nonnegative cost deductions and flat equity verified

Performance, risk and code review

LAB 6

Use AI to build and review performance, risk and code review; demonstrate the result with executable evidence.

You'll build

metrics.json and risk.png

Tools: Python · AI coding assistant · synthetic dataset

Performance, risk and code review

Test it

known drawdown fixture equals -25%

Recap — Building and Backtesting Strategies with AI

1

You can now: LO3: Backtest with execution delay and costs, and interpret return and risk.

TOPIC 04

Automating and Running Trading with AI

Parameter validation · broker API · monitoring

KEY CONCEPTS

Key Concepts — Automating and Running Trading with AI

1

Choose parameters on training data; reserve an untouched time holdout.

2

Compare parameter neighbourhoods rather than one winning setting.

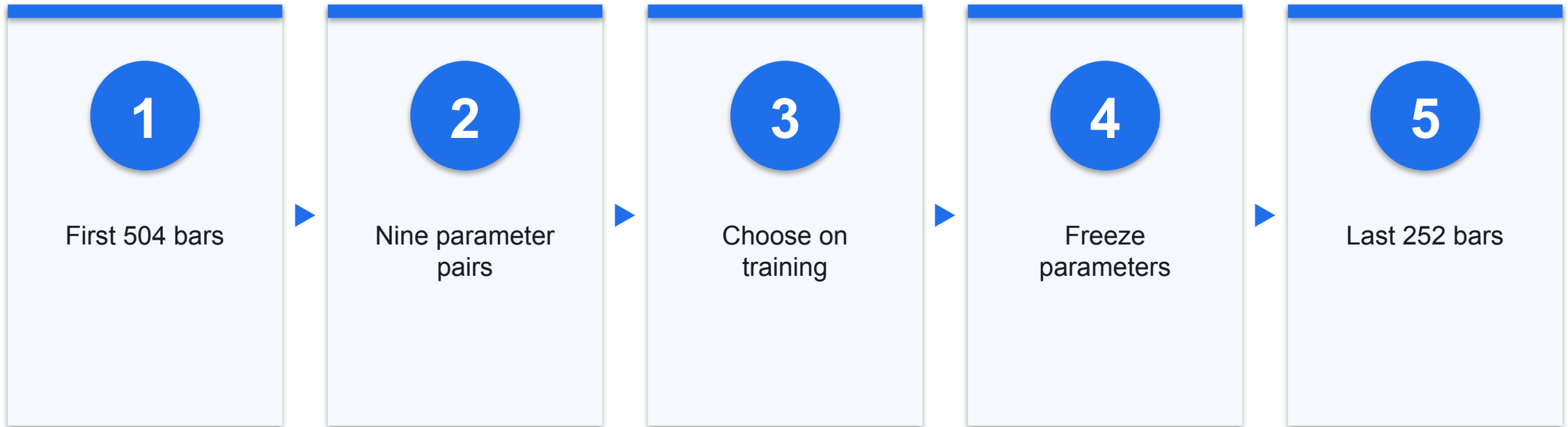
3

A broker adapter separates research from order execution.

4

Monitor stale data, duplicate requests, rejected orders and kill switches.

Chronological validation



Optimisation traps

1

Leakage

Random time-series split mixes past and future.

2

Selection bias

Trying more settings can fit noise.

3

Instability

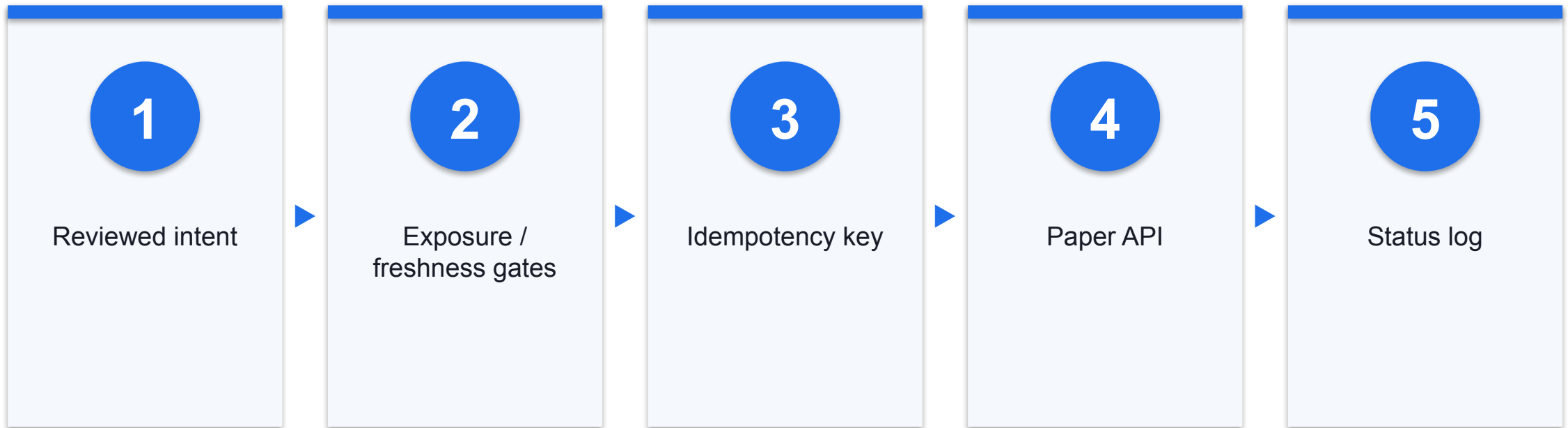
An isolated winning pair is fragile.

4

Discipline

Record all candidates; inspect neighbouring settings.

Broker adapter boundary



Paper API: distinct operations

1

Connection

Read-only GET /v2/account on the paper endpoint.

2

Submission

A separate operation; not implemented in sample.py.

3

Order state

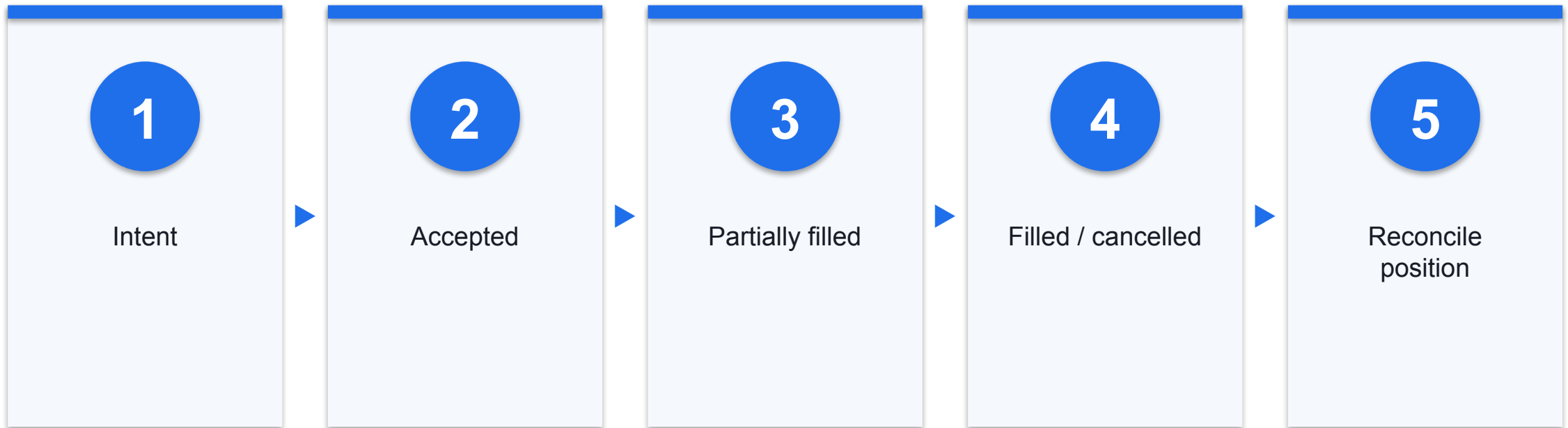
Accepted does not mean filled; track lifecycle.

4

Secrets

Paper keys in environment variables; never in prompts.

An order has a lifecycle



Monitor and stop

1

Stale data

Block intent when input freshness expires.

2

Duplicate retry

Use a stable id; reconcile before resubmitting.

3

Exposure breach

Reject quantity outside a declared cap.

4

Kill switch

Block new orders, log state and require review.

Research handover

1

Reproduce

Data source, dependency versions and configuration.

2

Explain

Signal logic, execution assumption and costs.

3

Observe

Logs, failures, drift and reconciliation.

4

Improve

One reviewed change at a time; rerun fixtures.

WHAT YOU'LL DO

Hands-On Labs — Automating and Running Trading with AI

Lab 7

- Optimise without leaking the future

Lab 8

- Broker API and monitored paper bot

Optimise without leaking the future

LAB 7

Use AI to build and review optimise without leaking the future; demonstrate the result with executable evidence.

You'll build

training-grid.csv and holdout.json

Tools: Python · AI coding assistant · synthetic dataset

Optimise without leaking the future

Test it

9 training candidates and 252 holdout observations

Broker API and monitored paper bot

LAB 8

Use AI to build and review broker api and monitored paper bot; demonstrate the result with executable evidence.

You'll build

events.jsonl

Tools: Python · AI coding assistant · synthetic dataset

Broker API and monitored paper bot

Test it

duplicate, stale-data, exposure-limit and halt events blocked

Recap — Automating and Running Trading with AI

1

You can now: LO4: Validate parameters chronologically and operate a monitored paper-trading workflow.



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Prompt and review

Specify, generate and validate Python trading code with an AI assistant.

2

Data and signals

Validate price data, visualise indicators and construct causal signals.

3

Backtest and risk

Backtest with execution delay and costs, and interpret return and risk.

4

Validate and monitor

Validate parameters chronologically and operate a monitored paper-trading workflow.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Your course material stays available at <https://lms-tms.tertiaryinfotech.com/>.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

THANK YOU FOR ATTENDING

Thank You!

You now have the practical skills from every lab in this course — keep applying them to your own work.